Case Study 8: Student Group Formation with Score Balancing

Aim: To develop a Python program that arranges students into groups based on their scores and specified group sizes while minimizing differences between group totals.

1. Problem Statement

A university wants to divide students into different groups for academic activities and projects. Each student has an individual score. The groups should be formed so that the total scores of all groups are as balanced as possible.

The program should:

1. Accept student names and scores.

2. Accept the required number of groups.

3. Arrange students according to their scores.

4. Allocate students into groups of specified sizes.

5. Calculate the total score of each group.

6. Minimize differences between group totals.

7. Display the final groups and their scores.

8. Assumed Conditions

For this program:

- Students are arranged in descending order of scores.

- Students are allocated one by one to the group with the lowest current total.

- If two groups have equal totals, the group with fewer students is selected.

- If both totals and group sizes are equal, the group with the smaller group number is selected.

3. Algorithm

4. Start.

5. Enter the number of students.

6. Enter student names and scores.

7. Enter the number of groups.

8. Enter the size of each group.

9. Sort students according to descending scores.

10. Initialize empty groups and group totals.

11. Select the student with the highest score.

12. Allocate the student to the group with the lowest total score.

13. Repeat until all students are allocated.

14. Calculate the total score of each group.

15. Display group members and their scores.

16. Stop.

17. Pseudocode

```
START

INPUT number_of_students

CREATE empty student list

FOR each student

    INPUT student_name
    INPUT student_score

    STORE student details

END FOR

INPUT number_of_groups
```

INPUT size of each group

SORT students by descending score

CREATE empty groups

INITIALIZE group totals to zero

FOR each student

    FIND group with minimum total score

    CHECK group size limit

    ADD student to selected group

    UPDATE group total

END FOR

DISPLAY group members

DISPLAY group totals

DISPLAY score difference

STOP

5. Python Program

Complete Python solution

```
# Student Group Formation with Score Balancingstudents = []# Input student detailsn = int(input("Enter number of students: "))for i in range(n):    print("\nStudent", i + 1)    name = input("Enter student name: ")    score = int(input("Enter student score: "))    students.append((name, score))# Input group detailsgroups_count = int(input("\nEnter number of groups: "))group_sizes = []for i in range(groups_count):    size = int(input("Enter size of group " + str(i + 1) + ": "))    group_sizes.append(size)# Validate group sizesif (groups_count <= 0 or    any(size <= 0 for size in group_sizes) or    sum(group_sizes) != n):    print("Invalid group sizes!")
```

6. Sample Output

Enter number of students: 6

Student 1

Enter student name: Ravi

Enter student score: 90

Student 2

Enter student name: Priya

Enter student score: 85

Student 3

Enter student name: Arun

Enter student score: 80

Student 4

Enter student name: Sneha

Enter student score: 75

Student 5

Enter student name: Kiran

Enter student score: 70

Student 6

Enter student name: Divya

Enter student score: 65

Enter number of groups: 2

Enter size of group 1: 3

Enter size of group 2: 3

====== STUDENT GROUP RESULTS ======

Group 1

Ravi - 90

Arun - 80

Divya - 65

Group Total: 235

Group 2

Priya - 85

Sneha - 75

Kiran - 70

Group Total: 230

====== FINAL SUMMARY ======

Highest Group Score: 235

Lowest Group Score: 230

Difference Between Groups: 5

7. Flowchart

#chatgpt-mermaid-_r_2b8_{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;fill:rgb(237, 237, 237);}@keyframes edge-animation-frame{from{stroke-dashoffset:0;}}@keyframes dash{to{stroke-dashoffset:0;}}#chatgpt-mermaid-_r_2b8_ .edge-animation-slow{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 50s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_2b8_ .edge-animation-fast{stroke-dasharray:9,5!important;stroke-dashoffset:900;animation:dash 20s linear infinite;stroke-linecap:round;}#chatgpt-mermaid-_r_2b8_ .error-icon{fill:rgb(48, 48, 48);}#chatgpt-mermaid-_r_2b8_ .error-text{fill:rgb(237, 237, 237);stroke:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2b8_ .edge-thickness-normal{stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ .edge-thickness-thick{stroke-width:3.5px;}#chatgpt-mermaid-_r_2b8_ .edge-pattern-solid{stroke-dasharray:0;}#chatgpt-mermaid-_r_2b8_ .edge-thickness-invisible{stroke-width:0;fill:none;}#chatgpt-mermaid-_r_2b8_ .edge-pattern-dashed{stroke-dasharray:3;}#chatgpt-mermaid-_r_2b8_ .edge-pattern-dotted{stroke-dasharray:2;}#chatgpt-mermaid-_r_2b8_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2b8_ .marker.cross{stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2b8_ svg{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:16px;}#chatgpt-mermaid-_r_2b8_ p{margin:0;}#chatgpt-mermaid-_r_2b8_ .label{font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2b8_ .cluster-label text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2b8_ .cluster-label span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2b8_ .cluster-label span p{background-color:transparent;}#chatgpt-mermaid-_r_2b8_ .label text,#chatgpt-mermaid-_r_2b8_ span{fill:rgb(237, 237, 237);color:rgb(237, 237, 237);}#chatgpt-

mermaid-_r_2b8_ .node rect,#chatgpt-mermaid-_r_2b8_ .node circle,#chatgpt-mermaid-_r_2b8_ .node ellipse,#chatgpt-mermaid-_r_2b8_ .node polygon,#chatgpt-mermaid-_r_2b8_ .node path{fill:rgb(9, 23, 44);stroke:rgb(31, 78, 148);stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ .rough-node .label text,#chatgpt-mermaid-_r_2b8_ .node .label text,#chatgpt-mermaid-_r_2b8_ .image-shape .label,#chatgpt-mermaid-_r_2b8_ .icon-shape .label{text-anchor:middle;}#chatgpt-mermaid-_r_2b8_ .node .katex path{fill:#000;stroke:#000;stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ .rough-node .label,#chatgpt-mermaid-_r_2b8_ .node .label,#chatgpt-mermaid-_r_2b8_ .image-shape .label,#chatgpt-mermaid-_r_2b8_ .icon-shape .label{text-align:center;}#chatgpt-mermaid-_r_2b8_ .node.clickable{cursor:pointer;}#chatgpt-mermaid-_r_2b8_ .root .anchor path{fill:rgb(175, 175, 175)!important;stroke-width:0;stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2b8_ .arrowheadPath{fill:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2b8_ .edgePath .path{stroke:rgb(175, 175, 175);stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ .flowchart-link{stroke:rgb(175, 175, 175);fill:none;}#chatgpt-mermaid-_r_2b8_ .edgeLabel{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_2b8_ .edgeLabel p{background-color:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2b8_ .edgeLabel rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2b8_ .labelBkg{background-color:rgba(0, 0, 0, 0.5);}#chatgpt-mermaid-_r_2b8_ .cluster rect{fill:rgb(48, 48, 48);stroke:rgba(255, 255, 255, 0.15);stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ .cluster text{fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2b8_ .cluster span{color:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2b8_ div.mermaidTooltip{position:absolute;text-align:center;max-width:200px;padding:2px;font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";font-size:12px;background:rgb(48, 48, 48);border:1px solid rgba(255, 255, 255, 0.15);border-radius:2px;pointer-events:none;z-index:100;}#chatgpt-mermaid-_r_2b8_ .flowchartTitleText{text-anchor:middle;font-size:18px;fill:rgb(237, 237, 237);}#chatgpt-mermaid-_r_2b8_ rect.text{fill:none;stroke-width:0;}#chatgpt-mermaid-_r_2b8_ .icon-shape,#chatgpt-mermaid-_r_2b8_ .image-shape{background-color:rgb(0, 0, 0);text-align:center;}#chatgpt-mermaid-_r_2b8_ .icon-shape p,#chatgpt-mermaid-_r_2b8_ .image-shape p{background-color:rgb(0, 0, 0);padding:2px;}#chatgpt-mermaid-_r_2b8_ .icon-shape .label rect,#chatgpt-mermaid-_r_2b8_ .image-shape .label rect{opacity:0.5;background-color:rgb(0, 0, 0);fill:rgb(0, 0, 0);}#chatgpt-mermaid-_r_2b8_ .label-icon{display:inline-block;height:1em;overflow:visible;vertical-align:-0.125em;}#chatgpt-mermaid-_r_2b8_ .node .label-icon path{fill:currentColor;stroke:revert;stroke-width:revert;}#chatgpt-mermaid-_r_2b8_ .node .neo-node{stroke:rgb(31, 78, 148);}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].node rect,#chatgpt-mermaid-_r_2b8_ [data-look="neo"].cluster rect,#chatgpt-mermaid-_r_2b8_ [data-look="neo"].node polygon{stroke:url(#chatgpt-mermaid-_r_2b8_-

gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].swimlane.cluster rect{filter:none;}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].node path{stroke:url(#chatgpt-mermaid-_r_2b8_-gradient);stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].node .outer-path{filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].node .neo-line path{stroke:rgb(31, 78, 148);filter:none;}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].node circle{stroke:url(#chatgpt-mermaid-_r_2b8_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].node circle .state-start{fill:#000000;}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].icon-shape .icon{fill:url(#chatgpt-mermaid-_r_2b8_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2b8_ [data-look="neo"].icon-shape .icon-neo path{stroke:url(#chatgpt-mermaid-_r_2b8_-gradient);filter:drop-shadow(1px 2px 2px rgba(185,185,185,1));}#chatgpt-mermaid-_r_2b8_ .node text{font-size:14px;font-weight:600;letter-spacing:normal;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_2b8_ .edgeLabels text{font-size:13px;font-weight:600;letter-spacing:-0.08px;fill:rgb(153, 206, 255);}#chatgpt-mermaid-_r_2b8_ .node tspan[font-weight="normal"],#chatgpt-mermaid-_r_2b8_ .edgeLabels tspan[font-weight="normal"]{font-weight:600;}#chatgpt-mermaid-_r_2b8_ .edgeLabel .label rect{opacity:1;rx:13px;ry:13px;fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ .node rect,#chatgpt-mermaid-_r_2b8_ .node circle,#chatgpt-mermaid-_r_2b8_ .node ellipse,#chatgpt-mermaid-_r_2b8_ .node polygon,#chatgpt-mermaid-_r_2b8_ .node path{fill:rgb(0, 40, 77);stroke:rgba(255, 255, 255, 0.1);stroke-width:1px;}#chatgpt-mermaid-_r_2b8_ .node rect{rx:16px;ry:16px;}#chatgpt-mermaid-_r_2b8_ .mermaid-decision .label-container{fill:rgb(0, 14, 26);stroke:rgb(26, 62, 95);stroke-dasharray:2,2;}#chatgpt-mermaid-_r_2b8_ .edgePaths .flowchart-link{stroke:rgb(175, 175, 175);stroke-width:1px;stroke-linecap:round;stroke-linejoin:round;}#chatgpt-mermaid-_r_2b8_ .marker{fill:rgb(175, 175, 175);stroke:rgb(175, 175, 175);}#chatgpt-mermaid-_r_2b8_ :root{--mermaid-font-family:-apple-system-body,ui-sans-serif,-apple-system,system-ui,"Segoe UI",Helvetica,"Apple Color Emoji",Arial,sans-serif,"Segoe UI Emoji","Segoe UI Symbol";}StartEnter student names andscoresEnter number and sizes ofgroupsGroup sizes valid?Display invalid group sizesSort students by descendingscoresCreate empty groups andtotalsSelect next studentFind eligible group with lowesttotalAllocate student to groupUpdate group totalMore students?Display group members andtotalsCalculate score differenceStopNoYesYesNo

8. Concepts Used from Syllabus

Syllabus concept	Application
Variables	Student name and score
Data types	Integer and string
Operators	Addition and comparison
Conditional statements	Group selection
Loops	Student allocation
Lists	Storing students and groups
Nested lists	Storing multiple groups
Tuples	Student name and score
Tuple unpacking	Extracting student details
Sorting	Arranging students by scores
Algorithms	Balancing group totals

9. Conclusion

The Student Group Formation program allocates students into groups based on their scores and specified group sizes. It uses a score-balancing method to reduce differences between group totals. This program helps universities create balanced student groups for academic activities and projects.

Note: This program uses a greedy allocation algorithm. It attempts to balance group scores but does not guarantee the mathematically optimal grouping in every situation.